Naming Conventions

Shop360Bike Medallion Data Warehouse

Microsoft SQL Server | Olamide Akanni

This document explains how we name schemas, tables, views, columns, procedures, and script files across the warehouse. It covers the conventions we actually use and notes three places where we've made deliberate exceptions.

General Principles

- Case: snake_case throughout. Lowercase with underscores between words.
- Language: English for all object names.
- Reserved words: never used as object names.
- Abbreviations: kept out of table and column names, but used in procedure names (layer prefixes shortened to brz and slv).
- Plurality: dimensions and facts use the plural form of the business entity, like dim_customers.

Schema Naming

Four schemas, one for each medallion layer plus an orchestration layer:

Schema	Purpose	Contains
bronze	Raw data exactly as received from the source	Tables and load procedures
silver	Cleansed and standardized data	Tables and load procedures
gold	Business-ready star schema	Views and one generated table
dbo	Pipeline orchestration and logging	Tables and procedures

We use dbo for orchestration rather than a separate schema because logging needs to touch all layers. It sits outside the medallion pattern and is used by all of them.

Table Naming Conventions

Bronze Layer

Bronze tables preserve the original source system name followed by the table name, unchanged. This keeps the answer to "what did the source send us?" always visible in the schema.

Pattern: <source_system>_<entity>

Table	Source	Original Name
crm_cust_info	CRM (CSV)	cust_info.csv
crm_prd_info	CRM (CSV)	prd_info.csv
crm_sales_details	CRM (CSV)	sales_details.csv
erp_cust_az12	ERP (CSV)	CUST_AZ12.csv
erp_loc_a101	ERP (CSV)	LOC_A101.csv
erp_px_cat_g1v2	ERP (CSV)	PX_CAT_G1V2.csv
inventory	BikeShopOLTP	dbo.inventory
fx_rates	exchangerate-api.com	REST response
web_events	Clickstream	Generated events

Exception 1: The last three tables have no source prefix. By strict convention they'd be oltp_inventory, api_fx_rates, and web_web_events. We dropped the prefix because each source contributes exactly one table, making the prefix unnecessary noise, and web_web_events just looks wrong.

Silver Layer

Silver mirrors bronze exactly, using the same names. This makes the lineage from bronze to silver obvious and lets you trace a transformation by comparing two tables with the same name in different schemas. Renaming happens only when we reach gold, where source-style names become business-style names.

Gold Layer

Gold names are business-facing and use a prefix describing what the object does in the model.

Pattern: <category>_<business_entity>

Prefix	Meaning	Examples
dim_	Dimension	dim_customers, dim_products, dim_currency, dim_date
fact_	Fact table	fact_sales, fact_inventory, fact_fx_rates
vw_	Helper view	vw_fx_rates_readable
report_	Pre-aggregated report table	(not yet used)

Exception 2: We added the vw_ prefix as a fourth category. It marks objects in gold that sit outside the star schema, like vw_fx_rates_readable which exposes FX rates with currency codes instead of surrogate keys for ad-hoc queries. This keeps the star schema unambiguous.

Note: dim_date and dim_currency are both singular (not plural). dim_date follows universal convention, and dim_currency sounds better than dim_currencies with its ISO code reference.

Column Naming Conventions

Surrogate Keys

Every dimension primary key uses the _key suffix and is generated in gold with ROW_NUMBER(). Facts reference the surrogate key, never the source ID, so changes to a source's numbering scheme can't break warehouse relationships.

Pattern: <entity>_key

Column	Defined In	Used By
customer_key	dim_customers	fact_sales, fact_web_events
product_key	dim_products	fact_sales, fact_inventory
currency_key	dim_currency	fact_fx_rates (twice)
date_key	dim_date	fact_inventory, fact_fx_rates

When a fact table references the same dimension in multiple roles, the key name includes the role. For example, fact_fx_rates has base_currency_key and target_currency_key, both pointing to dim_currency.

Exception 3: fact_sales uses actual DATE values (order_date, shipping_date, due_date) instead of integer date keys. It joins dim_date on full_date rather than date_key. This breaks the surrogate key pattern used by the other facts, and we've added creating those keys to the roadmap.

Natural Keys

When a dimension includes the source system identifier, the column keeps its original name with no suffix. dim_customers has customer_key alongside customer_id; dim_products has product_key, product_id, and product_number. No suffix means it's a natural key.

Business Columns

- Named for what the business calls them, not the source. cst_firstname in silver becomes first_name in gold.
- No source system abbreviations in gold. Prefixes like cst_, prd_, and sls_ are dropped at the gold boundary.
- Boolean columns get the is_ prefix: is_weekend.
- Dates use the _date suffix (full_date is the exception, named for its role as the complete date alongside date_key).

Technical Columns

System-generated metadata uses the dwh_ prefix to keep technical columns separate from business ones.

Column	Type	Purpose
dwh_create_date	DATETIME2	Timestamp when the row was loaded into silver. On all silver tables.

Stored Procedure Naming

Layer Batch Loaders

Pattern: load_<layer>

Loads every table in a layer that comes in the CRM and ERP batch.

Procedure	Loads
load_bronze	All six CRM and ERP tables from CSV
load_silver	All six CRM and ERP tables from bronze

Per-Source Loaders

Pattern: load_<layer_abbrev>_<entity>

Loads a single source arriving outside the CSV batch. We abbreviate the layer to brz or slv to keep names readable.

Procedure	Loads
load_brz_inventory	Stock snapshots from BikeShopOLTP
load_slv_inventory	Stock snapshots from bronze
load_slv_fx_rates	Exchange rates from bronze
load_slv_web_events	Clickstream events from bronze

We use abbreviations because load_bronze_inventory reads awkwardly next to load_bronze, and dropping the layer entirely gives two procedures called load_inventory in different schemas. Abbreviating is the cleanest option, but it means these two patterns aren't interchangeable.

Orchestration Procedures

Orchestration procedures are verb phrases describing what they do. They don't load anything themselves, so load_ would be misleading.

Procedure	Does
log_and_run	Executes a procedure, times it, and logs the result
run_full_pipeline	Runs all load procedures in order with one run ID

Script File Naming

SQL Files

SQL files use a prefix describing what they create:

Prefix	Contains	Examples
ddl_	CREATE TABLE/VIEW	ddl_bronze.sql, ddl_silver.sql, ddl_gold_views.sql
proc_	CREATE PROCEDURE	proc_load_brz_crm_erp.sql, proc_run_full_pipeline.sql
quality_checks_	Validation queries	quality_checks_silver.sql, quality_checks_gold.sql

Python Scripts

Python files use verb_noun naming, describing what they do:

Script	Action
fetch_fx_rates.py	Gets daily rates from the API into bronze
generate_web_events.py	Creates synthetic clickstream events
send_slack_alert.py	Posts pipeline results to Slack

Note: The external source DDL files (ddl_b_ext_source.sql and ddl_s_ext_source.sql) abbreviate the layer to a single letter. They could be renamed to ddl_bronze_ext_source.sql and ddl_silver_ext_source.sql to match procedure naming, but we've left them as a known inconsistency.